Software Requirements Specification

Foodies Paradise — Recipe Discovery Platform · Version 1.0

1. Introduction

Foodies Paradise is a web application where users discover recipes, search by the ingredients they already have, save favourites and manage a personal profile. This document lists the functional and non-functional requirements for version 1.0.

2. User Registration

REG-01 A visitor shall register with full name, email address, password and password confirmation.

REG-02 The full name shall contain 2 to 50 characters, letters and spaces only.

REG-03 The email must be in a valid format and must not already be registered. If it is, show "An account with this email already exists."

REG-04 The password must be 8 to 20 characters long and contain at least one uppercase letter, one lowercase letter, one digit and one special character.

REG-05 Password and confirmation must match; otherwise show "Passwords do not match."

REG-06 On successful registration, the system shall send a welcome email within 2 minutes and redirect the user to the login page with the message "Registration successful. Please log in."

3. Login and Session

LOG-01 A registered user shall log in with email and password.

LOG-02 On invalid credentials, show "Invalid email or password" without revealing which field was wrong.

LOG-03 After 5 consecutive failed attempts, the account shall be locked for 15 minutes.

LOG-04 The session shall expire after 30 minutes of inactivity, returning the user to the login page.

LOG-05 A "Logout" button shall end the session; pressing the browser Back button afterwards must not show protected pages.

LOG-06 Email matching at login shall be case-insensitive.

4. Recipe Search

SRC-01 Users shall search recipes by name using a search bar; results appear as cards with image, title, cuisine and cooking time.

SRC-02 Users shall search by ingredients by entering 1 to 10 ingredients; recipes are ranked by how many entered ingredients they use.

SRC-03 Filters shall be available for cuisine (Indian, Italian, Chinese, Mexican, Continental) and maximum preparation time (15, 30, 60 minutes or Any). Filters can be combined with a search.

SRC-04 If no recipe matches, show "No recipes found. Try fewer ingredients or different filters."

SRC-05 The search query shall be trimmed of leading/trailing spaces and must be at most 100 characters. An empty search shows all recipes.

SRC-06 Results shall be paginated with 12 recipes per page.

5. Favourites

FAV-01 Logged-in users can mark a recipe as favourite using a heart icon; clicking again removes it.

FAV-02 A user can save at most 200 favourites; attempting more shows "Favourites limit reached."

FAV-03 Guests who click the heart icon shall be redirected to the login page.

6. Profile Management

PRF-01 Users can edit their full name and dietary preference (Vegetarian, Vegan, Non-vegetarian, Jain).

PRF-02 Users can upload a profile photo in JPG or PNG format, maximum 2 MB. Other formats or larger files show "Please upload a JPG or PNG image under 2 MB."

PRF-03 Recommendations on the home page shall match the user's dietary preference.

PRF-04 Users can change their password by entering the current password and a new password meeting REG-04. The new password must differ from the current one.

7. Non-Functional Requirements

NFR-01 Search results shall load within 2 seconds for up to 500 concurrent users.

NFR-02 Passwords shall be stored hashed (bcrypt); they must never appear in logs or API responses.

NFR-03 The site shall work on the latest versions of Chrome, Firefox and Edge, and be usable on screens from 360 px wide.

NFR-04 All form inputs shall be protected against SQL/NoSQL injection and cross-site scripting (XSS).